Taking the multiplication in place of addition and n^{th} root in place of division by n in the definition of arithmetic mean we get the definition of geometric mean.

Definition 5.4

Let n be any positive integer. Let $a_1, a_2, a_3, \ldots, a_n$ be n non-negative numbers. Then the number

$$\sqrt[n]{a_1 a_2 a_3 \ldots a_n}$$

is called the ***geometric mean*** of the numbers $a_1, a_2, a_3, \ldots, a_n$.

Here also the numbers $a_1, a_2, a_3, \ldots, a_n$ need not be distinct but it is necessary that the numbers are non-negative. The geometric mean of the numbers $4, 6, 9$ is $\sqrt[3]{216} = 6$. The arithmetic mean of these three numbers is $\frac{19}{3} = 6\frac{1}{3}$. Observe that the arithmetic mean is greater than the geometric mean in this case. Is this true always?

It can be proved that "*For any set of n non-negative numbers, the arithmetic mean is greater than or equal to the geometric mean*". That is, if AM denotes the arithmetic mean and GM denotes the geometric mean, then $AM \geq GM$.

Let us prove this inequality $AM \geq GM$ for two non-negative numbers.

Theorem 5.2: If AM and GM denote the arithmetic mean and the geometric mean of two nonnegative numbers, then $AM \geq GM$. The equality holds if and only if the two numbers are equal.

Proof. Let a and b be any two nonnegative numbers. Then

$$AM = \tfrac{a+b}{2} \quad \text{and} \quad GM = \sqrt{ab}.$$

We have, $(a+b)^2 - 4ab = (a-b)^2 \geq 0$ Thus, $(a+b)^2 - 4ab \geq 0$ which gives $(a+b) \geq 2\sqrt{ab}$. Hence $\frac{a+b}{2} \geq \sqrt{ab}$.
In other words, $AM \geq GM$.

Moreover, the equality holds if and only if $(a+b)^2 - 4ab = 0$. This holds if and only if $(a-b)^2 = 0$ which holds if and only if $a = b$. Thus $AM = GM$ if and only if $a = b$. ■

Geometrical Proof for $AM \geq GM$

Let a and b be any two nonnegative real numbers. If at least one of them is zero, then GM is 0 and hence we have nothing to prove. So let us assume that $a > 0$ and $b > 0$. We draw a straight line segment AB of length $a + b$ and a semi-circle having AB as diameter. Let M be the midpoint of AB. Then M is the center of the semi-circle drawn. Since M is the midpoint of AB, we have $AM = MB = \frac{a+b}{2}$. So the radius of the circle is $\frac{a+b}{2}$. Let D be the point on AB so that $AD = a$; then $DB = b$.

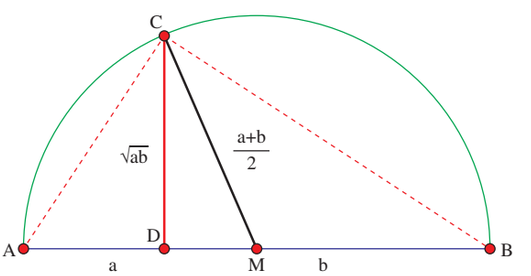


Figure 5.2